

Build a Polymer

Counting the bonds, the waters and the atoms in a growing chain — and finding the one macromolecule the counting rule does not fit.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/build-a-polymer/

1 Before you open anything

Answer from what you already know, and commit it to paper before you look at the demonstration. The point is to have something specific to compare against.

PREDICTION

A cell links 12 glucose monomers into a single straight chain of starch. How many water molecules are released in building that chain?

Write one sentence saying how you arrived at that number.

2 Read the counters

Open the demonstration. The **Carbohydrate** tab should already be selected; if not, click it. The **Chain length** slider runs from 1 to 12 in whole monomers, and the Counters panel reports the molecule four ways.

1. Click **Reset**, then drag **Chain length** down to **1**.
2. Record the readouts in the first row of the table below. **Formula** and **Molar mass** are in the Counters panel; copy the formula exactly, subscripts included.
3. Move the slider to 2, then 3, then 4, filling in a row each time.

Chain length	Bonds	Waters out	Molar mass (Da)	Formula
1				
2				
3				
4				

a. Two of the columns you filled in are identical all the way down. Which two, and what does that tell you about the relationship between bond formation and water?

b. Copy the two values from the lower part of the Counters panel: **Bond formed** = _____ and **Water lost per bond** = _____ Da.

c. With the chain at 4, press **Hydrolyze** once. Which of the two rows on the equation card lights up, and which way does the H₂O pill travel — toward the chain or away from it?

d. Now press **Add monomer** once and answer the same two questions. State in one sentence what hydrolysis does to water that dehydration synthesis does not.

3 The counting rule

Vary one control — chain length — over its whole range and look for the pattern in the numbers, not in the picture.

1. Stay on the **Carbohydrate** tab.
2. Set **Chain length** to each value in the left column and record the three readouts. Let the animation finish before you read the numbers.

Monomers (<i>n</i>)	Bonds	Waters out	Molar mass (Da)
1			
2			
4			
8			
12			

a. Write the rule in your own words, then as an expression in n : a straight chain of n monomers contains _____ bonds and releases _____ waters.

b. Say why the two counts are one *less* than the number of monomers rather than equal to it. Refer to a specific monomer in the chain.

c. Go back to your prediction in Part 1. Does the demonstration agree with it? If not, name the specific assumption that turned out not to hold.

d. The note under the **Build** panel gives the mass of one α -D-glucose monomer as 180.16 Da. Twelve of them would be 2161.9 Da, but your table says the 12-unit chain is lighter than that. Show that the missing mass is exactly accounted for by the waters that left.

4 Backwards, and where the rule stops working

A question rarely hands you n . It hands you a mass, or a molecule that the rule was never built for.

a. A straight chain of α -D-glucose has a molar mass of 1153.0 Da. Using 180.16 Da per monomer and 18.02 Da per water, work out how many monomers it contains. Show your algebra, then set the slider to your answer and check the **Molar mass** readout.

1. Click the **Lipid** tab. The slider is relabelled **Fatty acids attached** and now runs 0 to 3.
2. Set it to **3** — a complete triglyceride — and fill in the row below. A fifth readout, **$n - 1$ says**, appears on this tab; record it too.

Fatty acids	Bonds	Waters out	$n - 1$ says	Molar mass (Da)
3				

b. The **Waters out** and **$n - 1$ says** readouts disagree. Which one is correct for a triglyceride, and what is the missing molecule that the $n - 1$ rule failed to count?

c. Applying $n - 1$ to fatty acids is a reasonable move — it works for every other tab. State the structural feature a molecule must have for $n - 1$ to apply, and say in one sentence why a triglyceride does not have it.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working on the lines. Then reopen the demonstration, check yourself, and note where any arithmetic went off.

1. A single-stranded DNA fragment is 9 nucleotides long, every one of them dAMP (331.22 Da). How many phosphodiester bonds does it contain, how many water molecules were released in building it, and what is its molar mass to the nearest whole Da? (Water = 18.02 Da.)

2. A polypeptide built entirely from glycine (75.07 Da) has a molar mass of 645.6 Da. How many amino acid residues does it contain, and how many peptide bonds?

3. An amylase completely hydrolyzes one starch fragment of 8 glucose units into 8 free glucose molecules. How many water molecules are consumed? Compare the combined mass of the products with the mass of the original fragment: which is greater, and by how many Da?

4. A triglyceride releases 3 water molecules when it forms, but a chain of 3 glucose monomers releases only 2. Explain the difference in terms of what is being joined to what in each case. Do not simply restate the two rules.

Every mass on this page comes from one specific monomer per family — α -D-glucose, glycine, dAMP, palmitic acid — so that the numbers can be checked by hand. A real protein mixes 20 different amino acids and real DNA mixes four nucleotides, so an actual chain's mass is a sum of different residues, not a multiple of one.

6 On your own

Nothing here is graded. The demonstration holds more than this worksheet used.

- Switch between the four tabs with the equation card in view. Three of them show the same reaction; the **Protein** tab shows a different one. Work out which group hands over the hydrogen in each case, and why the peptide bond is the exception.
- On the **Lipid** tab, take the slider all the way down to **0**. Read the caption and the **Formula** readout for glycerol on its own, then add fatty acids one at a time and track which readouts change and which do not.
- The note under **Build** on the **Carbohydrate** tab says cellulose has the same formula and the same mass per unit as starch. If mass and formula cannot tell them apart, what does – and why can you not digest one of them?
- Read the last paragraph of “What this is for”, on how a cell actually builds DNA. Decide what the demonstration still gets right about a phosphodiester bond and what it leaves out about the way the cell pays for it.